

Estimation plan: parameters, identification, and moments

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Summary

This memo records the estimation as it now stands: which parameters are estimated, what identifies each one, which data moments are targeted, and the three places where the specification you gave needed a decision. It closes with the parameters left for the next step and what would identify them.

Everything below is implemented and the model solves. Nothing has been estimated yet under the new specification — that run comes after you have seen this.

1. What changed in the specification

Preference weights are time-invariant. `phi_1`, `phi_2`, `phi_3`, `lambda_1`, `lambda_2` were age profiles ($\text{phi_j0} + \text{phi_j1}*(t-1)$) held in per-period vectors. They are now scalars. Utility is defined only up to *relative* weights, so two of the five must be normalised: **`phi_1 = 1` and `lambda_1 = 1`**.

`psi_terminal = 0`. This is the weight on the child's human capital in the parent's terminal value at separation, $\text{psi}*\log(k) + \text{kappa}*\log(a)$. At zero the parent values the child's skill only through the flow term $\text{phi_3}*\log(\text{HC})$ and through altruism `omega`.

Worth knowing: `psi_terminal` had been raised $1.0 \rightarrow 4.0$ precisely because a low terminal weight made the last period value skill far less than every earlier one, and `tau_p` collapsed at $t = 17$. **I checked, and it does not recur.** At `psi = 0` the model solves and `tau_p` runs $0.570 \rightarrow 0.239$ at $t = 13 \rightarrow 0.314$ at $t = 17$. The flow term and altruism carry it.

Human capital is now in the data's units. *HC* is the Woodcock–Johnson PCA composite (a W-score, roughly 300–600), not an abstract index on $[0.001, 10]$. This was necessary — see §3.

Initial human capital comes from the data. `sim_hc_init` was `Uniform(0,1)`, which is arbitrary. It is now lognormal, fitted to the PCA and extrapolated back to age 0.

2. Estimated parameters and what identifies each

Nine parameters against ten moments.

Parameter	Identified by	Data moment
phi_2	parents' willingness to give up leisure for income	mean hours worked
phi_3	how much parents value the child's skill	parental time and monetary investment , jointly with the HC level
lambda_2	how much the <i>child</i> values its own skill	the child's own study time
R_0	the level of the HC technology (TFP)	the HC level
sigma_1_0	elasticity of HC to parental time	mean t_p , ages 1–9
sigma_1_1	its age slope	mean t_p , ages 10–17
sigma_2_0	elasticity of HC to money	mean e_p , ages 1–9
sigma_2_1	its age slope	mean e_p , ages 10–17
sigma_4_0	elasticity of HC to the child's own study	mean i_c , ages 6–17

Fixed, not estimated: phi_1 = 1 and lambda_1 = 1 (normalisations); sigma_3 (self-productivity, 0.407); mu_1 (see §3).

Why the human-capital moments are essential

Valuation and technology are otherwise collinear. phi_3 and lambda_2 say how much parent and child *value* skill; R_0, sigma_1, sigma_2, sigma_4 say how efficiently skill is *produced*. **Both raise investment.** Looking only at **t_p**, **e_p** and **i_c**, no combination of moments can separate them — the optimizer would sit on a ridge and return whatever its starting point was near.

What separates them is the resulting **HC level**: technology raises it, valuation does not. That moment only became usable once HC was placed in the data's units, which is why the rescaling in §1 was not cosmetic.

3. Three places the specification needed a decision

3.1 mu_1 cannot be estimated alongside sigma_4_1

Your reasoning — that mu_1 is identified from phi_3 and lambda_2, because only phi_3 operates before age 6 and all three after — is right in direction. The sharper statement is that with lambda_1 = 1 the child's effective weight on its own leisure is exactly (1 - mu_t), so **lambda_2 sets the level of study time and mu_1 sets its slope in age.**

But the child's study first-order condition is driven by the ratio

$$\frac{\sigma_{4,t}}{(1 - \mu_t) \lambda_1}, \quad \sigma_{4,t} = \exp(\sigma_{40} + \sigma_{41}(t - 1))$$

and sigma_41 and mu_1 both bend that ratio with age. **The study-time profile cannot identify both.** One must go.

I dropped **sigma_4_1** and kept **mu_1** fixed: **mu_1** is a welfare weight pinned by the model’s structure, whereas **sigma_4** is a technology parameter that can reasonably be held flat in age. **sigma_4** is therefore constant in **t**. If you would rather estimate **sigma_4_1**, **mu_1** must first be fixed by an explicit rule — and I would want that rule from you rather than guessing it.

3.2 Extrapolating the PCA back to age 0

The test is not administered before age 3, so both the mean and the SD of log HC are fitted log-linearly on ages 3–17 and extrapolated:

$$\text{mean log HC} = 5.9290 + 0.02392 \text{ age} \quad (R^2 = 0.835)$$

$$\text{sd log HC} = 0.0698 - 0.00310 \text{ age} \quad (R^2 = 0.638)$$

giving $\log \text{HC}_0 \sim \text{Normal}(5.9290, 0.0698)$ — a median of **376** and a 10–90 range of **344–411**. The simulated draw reproduces this (median 375.6, p10 344, p90 412).

Log-linear rather than quadratic, on two grounds: it is the same functional form the production function uses, and a quadratic — though it tracks ages 3–17 slightly better — bends to **303** at age 0 against the linear **376**, a 24% difference from extrapolating three years beyond its support.

3.3 The model over-produces human capital by ~50%

Putting HC in data units made a pre-existing error visible:

child age	1	5	9	13	17
model HC	376	815	753	717	803
data	385	423	466	513	564

This is not a rescaling artefact. The cascade faithfully preserved the old dynamics, and the old model grew HC by 5.6× where the data grows by 1.5×. The error was always present; it was invisible while HC had no units.

R_0 is the parameter that sets that level, so **it has been added to the estimated set** — nine parameters, still over-identified against ten moments.

4. The moments

All in the model’s own units: one model unit is \$10,000/year, time is a share of the 112-hour non-sleep week per parent, HC is a log W-score.

Moment	Source	Target	N
mean_c_p	cons_exhous_real_w99	3.1577	6,742
mean_h_p	(wh_mom + wh_dad)/2 / 112	0.3070	15,665
mean_t_p_early	par_time_tot / 112, ages 1–9	0.4672	475
mean_t_p_late	par_time_tot / 112, ages 10–17	0.3232	590
mean_e_p_early	m_method2_final_w99, ages 1–9	0.3532	8,178

Moment	Source	Target	N
mean_e_p_late	m_method2_final_w99, ages 10–17	0.4414	7,182
mean_i_c_early	study_hrs / 112, ages 6–9	0.0386	171
mean_i_c_late	study_hrs / 112, ages 10–17	0.0500	584
mean_hc_early	x_gach (log PCA), ages 3–9	6.0802	253
mean_hc_late	x_gach (log PCA), ages 10–17	6.2558	549

Two notes. The child’s study moment starts at **age 6**, not 1: there is no child decision before `T_CHILD_VOICE = 6`, so averaging in ages 1–5 would target a number the model cannot produce. And `t_p` and `e_p` are split early/late because a single mean cannot separate an age *slope* from a *level*.

A caveat that must travel with any estimate of `phi_2`

`t_p` is measured with `par_time_tot` — active **plus** nearby/supervisory presence. That measure does not fit an exhaustive time budget: per parent, **leisure + work + own active care = 112.00** exactly, but with `par_time_tot` it is **133.25**, twenty-one hours over. Since the model enforces `l_p + h_p + t_p = 1`, targeting `h_p` and `t_p` **forces** model leisure to about 33 hrs/wk against the 59 hrs/wk the same dataset measures. That gap lands in `phi_2`. **The estimated `phi_2` should not be read as a clean taste for leisure.** Switching the target to per-parent active time closes the identity exactly, if you prefer that.

5. Optimiser settings

TikTak (Arnoud, Guvenen & Kleineberg 2022), at the paper’s own ratio: **N = 1,000 Sobol’ points, N* = 100 restarts** (N* = 0.1N).

The Sobol stage is parallel (~8 minutes on 20 workers). The 100 restarts are **sequential by construction** — restart *j* starts from the best point found by 1..j-1 — so they dominate: budget roughly **25 hours**. If that is too long, cut restarts rather than Sobol points; more Sobol points are nearly free and give the local stage better seeds.

6. Next step: parameters not yet estimated

These are deliberately left for a second stage, with what would identify each.

Parameter	What it is	Would be identified by
<code>kappa_terminal</code>	weight on the parents’ retained assets in their terminal value	parental net worth after the child leaves — assets_real at child age 17, and post-separation wealth for families whose child has finished college
<code>kappa_0</code>	level of the psychic cost of college	the overall college enrolment rate

Parameter	What it is	Would be identified by
<code>kappa_theta</code>	ability gradient in the psychic cost (negative: ability lowers it)	the enrolment gradient in child test scores — enrolment by <code>g_ACH</code> quantile
<code>kappa_ParEd</code>	parental-education shift (negative)	the enrolment gap by parental education — <code>BothCollege</code> vs not

The three **kappa** terms of the psychic cost all move enrolment, so they need three *distinct* enrolment moments: the level, the slope in ability, and the gap by parental education. Estimating them against the overall rate alone would not identify them.

`kappa_terminal` is separate: it governs how much wealth parents retain rather than transfer, so it is identified by post-separation wealth, not by enrolment.

Their current values are transported from Colas (Table 2) by **sign and ratio only** — `kappa_ParEd/kappa_theta` = 0.205 is hers; the levels are not, because her utility is $c^{(1-\gamma)}/(1-\gamma)$ at $\gamma = 1.9$ with consumption in dollars while ours is $\rho = 1.5$ with c of order 0.1–5.

7. Open questions

1. **mu_1.** It is currently fixed at -0.04 , giving a parental weight of 0.52 at $t = 17$. Should it be pinned by an explicit rule — equal weights at 18, or zero parental weight at 18 — or estimated in place of `sigma_4_1`?
2. **t_p measure.** Keep `par_time_tot`, accepting that `phi_2` absorbs a 21-hour time-budget inconsistency, or switch to per-parent active time, which closes the identity exactly?
3. **Sample.** The targets are built from all one-child families. The model assumes parents aged 26 at birth, which matches the *cohort* file (father 25–30). The restriction moves consumption by -7.7% , monetary investment by -17.8% and assets by -28% , while time moments barely move. Should the targets be rebuilt on the cohort?